

Dynamic energy budget modelling of anuran life-history with account for individual variability in developmental rates[☆]

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ABSTRACT

The decline of amphibian populations is a major concern in environmental protection, and likely involves multiple stressors, such as loss of habitat, temperature changes, chemical exposure and pathogens. Mechanistic models can be helpful to understand and predict amphibian response to multiple stressors, but amphibian physiology and life history are unique, requiring adaptations to generic models.

Thus, the goal of this study was to implement and test a whole life-cycle Dynamic Energy Budget (DEB) model for amphibians and test its ability to reproduce and predict metamorphosis traits.

The model includes a dedicated reserve compartment for the biomass that is used to fuel metabolic processes during metamorphic climax, and a decline in feeding rates for metamorphosing individuals. While the model formulation was partially guided by a minimization of the number of parameters, a global sensitivity analysis indicated nevertheless that a wide range of metamorphosis traits could be produced.

The model reproduced larval growth and metamorphosis traits in *Discoglossus galganoi* with account for individual variability in the time to reach metamorphosis with high accuracy.

Based on the *D. galganoi* parameters as a reference, we fitted the model to a subset of the same life-history data for *Pelophylax perezi* larvae, attempting to predict the duration of metamorphosis and wet mass at the end of metamorphosis. For *P. perezi*, the duration of metamorphosis was slightly under-predicted, and the wet mass at the end of metamorphosis was under-predicted by a factor of 0.6. Given that *D. galganoi* and *P. perezi* are rather distinct in their developmental strategies and stem from different families of anurans, this result does show potential for possible model applications to anuran species with poorer data coverage, in particular if data from a more closely related reference species is available. Open questions remain with respect to implementing the fasting period during metamorphic climax in models based on DEBkiss, while maintaining the fit to metamorphic mass change data.

1. Introduction

1.1. Context & motivation

Amphibians are a taxonomic group in global decline, and the reasons are likely multi-causal. Stressors like chemical exposure, infectious diseases, alien species and land use change exert in combination pressure on amphibian populations (Collins and Storfer, 2003).

These stressors act within the context of a dynamic system, including the metabolic organization of the amphibian organism itself. To understand the effects of these stressors mechanistically, and possibly

predict combined effects, an important step is the formulation of mechanistically underpinned mathematical description of the amphibian life-cycle, in terms of a dynamic system.

1.2. Dynamic energy budget theory is a generic framework to model life history

A commonly applied theory to describe the life-history of organisms as a dynamic system is Dynamic Energy Budget (DEB) theory. DEB theory describes the uptake and allocation of energetic resources as an

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energy or mass balance, where resource can be primarily invested into maintenance, somatic growth, maturation and reproduction.

Two frequently used formulations of this model are referred to as *standard DEB model* (stdDEB, Kooijman (2010)) and *DEBkiss* (Jager et al., 2013). The latter is, historically, a simplification of stdDEB, but can also be derived from first principles. The standard DEB model includes a reserve compartment which buffers the use of assimilated energy, and de-couples somatic mass (structure) from short-term fluctuations in energy assimilation. The stdDEB reserve is therefore generally not pure storage compartment, but continuously used by the organism.

The DEBkiss formulation omits this reserve component and assumes that assimilates are used instantaneously. The state variable maturity, used to control life stage transitions when using body size is not appropriate, can optionally be included, and is then associated with its own maintenance costs.

Irrespective of the model formulation used, the primary outputs of a DEB model simulation are typically somatic growth and reproductive output over time and the rates of DEB processes driving these processes, such as assimilation rate and maintenance costs.

From this primary output, numerous observable traits can be derived, such as the organism state at life stage transitions, or the cumulative amount of resources invested in DEB processes. The underlying parameters are species-specific metabolic rates and efficiencies, which are in most cases inferred statistically from life history data. An appeal of DEB theory is that it summarizes some principles that apply to all animals. The underlying parameters can be used to extrapolate from observed patterns in life-history to unobserved scenarios. For specific cases, DEB models can be adapted in order to account for the biological idiosyncracies of specific organisms (e.g. Llandres et al. (2015)).

1.3. How can DEB theory be applied to amphibians?

Adaptions of generic DEB models are clearly necessary to model the amphibian life cycle, whose most obvious idiosyncrasy is the undergoing of metamorphosis. Metamorphosis represents a drastic reorganization of the organism (Zhu et al., 2020). In the simplest case, a model may view pre- and post-metamorphic stages as two separate DEB organisms, each with their own metabolic parameters. However, metamorphic climax is a critical phase during the amphibian life cycle and should be considered as a distinct developmental phase (Walsh, 2010).

During metamorphic climax, most of the morphological and physiological changes occur (Laurie Sanderson and Kupferberg, 1999). Ingestion rates decline and ingestion eventually stops completely. Most literature sources state that feeding stops at metamorphic climax (Hourdry et al., 1996; Duellman and Trueb, 1994). Residual feeding beyond metamorphic climax and beyond is reported rarely (Kuzmin, 1997) and might only extend for short periods beyond the development stage that mark metamorphic climax, i.e. Gosner stage 42 in anurans (Pfab et al., 2020). Most studies state that feeding stops at climax due to dysfunctionality of the feeding apparatus or digestive tract Hourdry et al. (1996), but usually without a precise account for the duration of the fasting period. Irrespective of the exact time-course in feeding patterns, the cessation or decline in feeding means that the metabolic processes of maintenance and maturation have to be supported by internal energy sources, requiring corresponding adaptations to DEB model formulations. For the development of a generic DEB model for amphibians however, a more precise temporal description of the change in feeding rates is needed, in particular to explain how, in some species, low decreases in dry mass may be observed (Beck and Congdon, 2003). Metamorphic climax can further vary in duration and the amount biomass lost due to consumption of internal energy sources. While the temporal pattern in wet mass often follows a clear decline during metamorphic climax, a significant part of this is due to changes in water content of the organism. In *Bufo terrestris*, a -35% change in wet mass during metamorphic climax, but no statistically significant change in

dry mass was observed (Beck and Congdon, 2003). Brown et al. (1988) demonstrated a relatively small decrease in dry mass during metamorphosis. In this study, we formulated a DEB model for amphibians based on DEBkiss. We aimed for a model that can be calibrated with data on growth and development of larval and metamorphic stages, and that can simultaneously reproduce a wide range of life-history traits. This necessitates the introduction of simplifying assumptions while retaining the necessary degree of model flexibility. We formulated a mathematical model and analysed the model's capacity to reproduce hypothetical and observed life-history traits in two species which differ in their developmental strategies. This lays the groundwork to apply this model for the analysis of environmental stressor effects and extrapolation to the population level.

2. Methods

2.1. Model description

The point of departure for model development was the DEBkiss model (Jager et al., 2013), with all state variables expressed in dry mass. DEBkiss has previously been used to describe amphibian metamorphosis (Pfab et al., 2020), and compared with a model that does include a reserve compartment according to stdDEB. A direct comparison indicated little difference between the DEBkiss and stdDEB variant, regarding fitted patterns of growth and respiration (Pfab et al., 2020). However, to model the entire life cycle under varying environmental conditions, we need additional flexibility to allow for plasticity in larval and metamorph development across environmental gradients.

The reserve in standard DEB (Kooijman, 2010) acts as a buffer rather than a storage compartment. The adaptation for the stdDEB model for amphibians thus also requires additional rules for the resource allocation during metamorphosis (Romoli et al., 2025). In combination, we see these aspects as a good indication that the full DEBkiss model as described by Jager et al. (2013), and including a maturity component, is a good starting point to model the amphibian life cycle.

Below, we first describe the formulation of the DEBkiss model used in the present study, and subsequently the extension for amphibians.

2.1.1. Baseline model

Firstly, we implemented a formulation of DEBkiss (Jager et al., 2013) that explicitly takes feeding into account and expresses all state variables in dry mass. For a generic animal, this means that the food ingestion rate $\dot{I}$ (mg d⁻¹) is expressed as:

$$\dot{I} = \frac{dI}{dt} = \{\dot{I}\}_{\max} f([X]) S^{2/3} \quad (1)$$

$\{\dot{I}\}_{\max}$ is the maximum size-specific ingestion rate (mg mg^{-2/3} d⁻¹) and S (mg) is structural mass. This means that we omit the usual conversion from structural mass to structural volume. Since the conversion is typically assumed to be linear and constant, omitting it has no consequence for the model dynamics, only for the dimension of $\{\dot{I}\}_{\max}$. As a consequence, the density of dry mass is not needed as a model parameter.

$f([X])$ is the scaled functional response and takes the form of a Holling type II functional response, depending on the resource density $[X]$ (mg L⁻¹) and the half-saturation constant K_X (mg L⁻¹):

$$f([X]) = \frac{[X]}{[X] + K_X} \quad (2)$$

The assimilation rate $\dot{A}$ follows from a constant assimilation efficiency η_{IA} (-).

$$\dot{A} = \eta_{IA} \dot{I} \quad (3)$$

Following the assumptions of DEB theory, the amount of available assimilates is divided into a branch κ for somatic growth ($\dot{S}$) and

maintenance ($\dot{M}$) and a branch $1 - \kappa$ for maturation ($\dot{H}$), maturity maintenance ($\dot{J}$) and reproduction ($\dot{R}$).

The structural growth rate $\dot{S}$ further depends on the allocation parameter κ . Somatic maintenance follows first-order kinetics depending on a maintenance rate constant k_M .

$$\dot{S} = \eta_{AS} (\kappa \dot{A} - \dot{M}) \quad (4)$$

$$\dot{M} = k_M S \quad (5)$$

The maturity state H triggers the transition to puberty, depending on the threshold H^p , and is associated with a maintenance rate $\dot{J}$. For adults ($H \geq H^p$), maturation stops and assimilates are instead invested into the reproduction buffer R , accounting for a reproduction efficiency η_{AR} .

$$\dot{H} = \begin{cases} (1 - \kappa)\dot{A} - \dot{J} & \text{if } H < H^p \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

$$\dot{R} = \begin{cases} 0 & \text{if } H < H^p \\ \eta_{AR} [(1 - \kappa)\dot{A} - \dot{J}] & \text{otherwise} \end{cases} \quad (7)$$

$$\dot{J} = k_J H \quad (8)$$

For the embryonic life stages, we again follow the heuristics of DEBkiss.

Embryos absorb nutrients from an internal resource X^{emb} at a rate $\dot{I}^{emb}$ at their maximum size-specific rate. Embryonic development is assumed to be completed when this resource is used up:

$$\dot{I}^{emb} = \begin{cases} \{I\}_{max}^{emb} S^{2/3} & \text{if } X^{emb} > 0 \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

Accounting for feedbacks between ingestion and resource, this leads to the following complete system of ODEs for the baseline model:

$$\dot{X} = \dot{X}_{in} - \dot{I} \quad (10)$$

$$\dot{X}^{emb} = -\dot{I}^{emb} \quad (11)$$

$$\dot{I} = \begin{cases} 0 & \text{if embryo} \\ \{I\}_{max} f([X]) S^{2/3} & \text{otherwise} \end{cases} \quad (12)$$

$$\dot{I}^{emb} = \begin{cases} \{I\}_{max}^{emb} S^{2/3} & \text{if embryo} \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

$$\dot{A} = \eta_{IA} \dot{I} \quad (14)$$

$$\dot{M} = k_M S \quad (15)$$

$$\dot{J} = k_J H \quad (16)$$

$$\dot{S} = \eta_{AS} (\kappa \dot{A} - \dot{M}) \quad (17)$$

$$\dot{H} = \begin{cases} (1 - \kappa)\dot{A} - \dot{J} & \text{if not adult} \\ 0 & \text{otherwise} \end{cases} \quad (18)$$

$$\dot{R} = \begin{cases} \eta_{AR} [(1 - \kappa)\dot{A} - \dot{J}] & \text{if adult} \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

The resource input rate $\dot{X}_{in}$ (mg d⁻¹) is treated as global parameter. By default, we assume *ad libitum* feeding conditions, which means that $f([X]) = 1$ is enforced.

The variables *embryo*, *juvenile* and *adult* are boolean life stage indicators:

$$embryo = \begin{cases} \text{true} & \text{if } X_{int}^{emb} > 0 \\ \text{false} & \text{otherwise} \end{cases} \quad (20)$$

$$juvenile = \begin{cases} \text{true} & \text{if } X_{int}^{emb} = 0 \wedge H < H^p \\ \text{false} & \text{otherwise} \end{cases} \quad (21)$$

$$adult = \begin{cases} \text{true} & \text{if } H \geq H^p \\ \text{false} & \text{otherwise} \end{cases} \quad (22)$$

2.1.2. Adaptations for amphibians

To adapt the model for the simulation of amphibian life-history, we needed to include additional life stages and define the dynamics of these life stages.

While the baseline model includes three life stages (embryo, juvenile, adult), the amphibian model includes five (embryo, larva, metamorph, juvenile, adult). Embryos and adults behave as in the baseline model.

To model larval development and their preparation of metamorphosis, we assumed that larvae accumulate a reserve E^{mt} . We define E^{mt} as the biomass which is mobilized during metamorphic climax in order to fuel metabolic processes. Importantly, this has a different purpose and definition than the reserve E in the standard DEB model (Kooijman, 2010). According to its definition, we consider the metamorphic reserve E^{mt} to be a functional compartment, rather than a specific tissue or compound. We assumed for the present analysis that it is not associated with maintenance costs, although we can imagine that this assumption could be weakened, in which case the interpretation of E^{mt} may be shifted towards a sub-compartment of structure.

We assumed that the allocation E^{mt} occurs within the κ -split and that the reserve allocation fraction is controlled by the parameter γ (–):

$$\dot{E}^{mt} = \eta_{AS} \gamma (\kappa \dot{A} - k_M S) \quad (23)$$

$$\dot{S} = \eta_{AS} (1 - \gamma) (\kappa \dot{A} - k_M S) \quad (24)$$

The formulation of Eq. (24) leads to an intuitive interpretation of γ , which is the relative contribution of E^{mt} to total biomass at equilibrium. For example, for $\gamma = 0.5$ and otherwise constant conditions, there will be equal amounts of E^{mt} and S at equilibrium.

The end of the larval period is triggered by a maturity threshold H^J . In the present study, we view this as equivalent to developmental stage Gosner 42 (Gosner, 1960), i.e. front limb emergence or when individuals reach metamorphic climax. Depending on the time-course of size-specific ingestion rates, an association to life stages as early as Gosner 38 might also be plausible. By definition, E^{mt} is used during metamorphosis to fuel mandatory processes, i.e. maintenance and maturation:

$$\dot{E}^{mt} = -(\dot{M} + \dot{J} + \dot{H}) \quad (25)$$

Eq. (25) implicitly assumes that E^{mt} is used with an efficiency of 100%. While this is generally not realistic, an additional conversion factor would not be identifiable from typical life history-data. An alternative would be to fix such a conversion factor to a default value below 100%. In lack of a clear default value, we omitted it for the present analysis. We assumed that ingestion rates decrease gradually during metamorphosis, limiting the fasting period to a small fraction of metamorphosis.

In order to calculate $\dot{E}^{mt}$ without introducing more parameters, we assumed that the κ rule applies to the mandatory fluxes during metamorphosis. In other words, energetic demands which definitely have to be covered are distributed according to standard DEB assumptions. These are taken from the biomass compartment (E^{mt}) which is, by earlier definition of the compartment, definitely available for consumption during metamorphosis:

$$\dot{H} = \frac{(1 - \kappa)\dot{M}}{\kappa} - \dot{J} \quad (26)$$

This temporarily decouples the dynamics of $\dot{S}$ from κ , which does not adhere to basic assumptions of DEB theory anymore. However, we think that this choice is justified, because metamorphosis represents such a drastic reorganization of the organism. We assume that metamorphosis is completed when E^{mt} is depleted, which is analogous to the DEBkiss rules for embryos and avoids the introduction of an additional parameter, which would be a maturity threshold to trigger the end of metamorphosis.

Simultaneously to the mobilization of reserves, ingestion rates decrease. To model such a decrease in a gradual manner without additional parameters, we introduced a relationship between the current size-specific ingestion rate $\{I\}_t^{mt}$ and the relative depletion of reserves:

$$\{I\}_t^{mt} = \{I\}_{max}^{lrv} \frac{E_t^{mt}}{E_{max}^{mt}} \quad (27)$$

Here, $\{I\}_t^{mt}$ is the dynamically calculated size-specific maximum ingestion rate at during metamorphosis, E_t^{mt} is the reserve level at the same time, and E_{max}^{mt} is the maximum reserve level, which is reached at the beginning of metamorphosis and tracked internally by the model simulation. Analogous to Eq. (1), multiplication of $\{I\}$ with $S^{2/3}$ yields the ingestion rate $\dot{I}_t^{mt}$.

With the mandatory fluxes $\dot{H}$, $\dot{M}$ and $\dot{J}$ covered by the reserve, the residual ingestion flux for metamorphs can only be linked to the facultative flux $\dot{S}_t^{mt}$ (accumulation of new structural mass at time t during metamorphic climax):

$$\dot{S}_t^{mt} = \eta_{AS} \dot{I} \quad (28)$$

Ingestion rates reach 0 for Gosner stage 46, corresponding to complete tail resorption, which marks the end of metamorphosis. This model behaviour is similar to the rules described by Pfab et al. (2020), where feeding rates were interpolated between Gosner stages 43 and 46.

Eqs. (28) and allow for a relatively long metamorphosis without significant changes in dry mass.

To summarize, the complete ODE system for the adapted amphibian model is then:

$$\dot{X}^{emb} = \begin{cases} -\dot{I} & \text{if embryo} \\ 0 & \text{otherwise} \end{cases} \quad (29)$$

$$\dot{X} = \begin{cases} 0 & \text{if embryo} \\ \dot{I} & \text{otherwise} \end{cases} \quad (30)$$

$$\dot{I} = \begin{cases} 0 & \text{if embryo} \\ \{I\}_{max} f([X]) S^{2/3} & \text{otherwise} \end{cases} \quad (31)$$

$$\dot{S} = \begin{cases} \eta_{AS}(1-\gamma)(\kappa\dot{A} - k_M S) & \text{if larva} \\ \eta_{AS}\dot{I}_t^{mt} & \text{if metamorph} \\ \eta_{AS}(\kappa\dot{A} - \dot{M}) & \text{otherwise} \end{cases} \quad (32)$$

$$\dot{E}^{mt} = \begin{cases} \eta_{AS}\gamma(\kappa\dot{A} - k_M S) & \text{if larva} \\ -(\dot{M} + \dot{J} + \dot{H}) & \text{if metamorph} \\ 0 & \text{otherwise} \end{cases} \quad (33)$$

$$\dot{H} = \begin{cases} 0 & \text{if adult} \\ (1-\kappa)\dot{A} - \dot{J} & \text{otherwise} \end{cases} \quad (34)$$

$$\dot{R} = \begin{cases} \eta_{AR} [(1-\kappa)\dot{A} - \dot{J}] & \text{if adult} \\ 0 & \text{otherwise} \end{cases} \quad (35)$$

The life stage indicators are extended to:

$$embryo = \begin{cases} \text{true} & \text{if } X_{int}^{emb} > 0 \\ \text{false} & \text{otherwise} \end{cases} \quad (36)$$

$$larva = \begin{cases} \text{true} & \text{if } X_{int}^{emb} = 0 \wedge H < H^j \\ \text{false} & \text{otherwise} \end{cases} \quad (37)$$

$$metamorph = \begin{cases} \text{true} & \text{if } H \geq H^j \wedge E^{mt} > 0 \\ \text{false} & \text{otherwise} \end{cases} \quad (38)$$

$$juvenile = \begin{cases} \text{true} & \text{if } H \geq H^j \wedge E^{mt} = 0 \wedge H < H^p \\ \text{false} & \text{otherwise} \end{cases} \quad (39)$$

$$adult = \begin{cases} \text{true} & \text{if } H \geq H^p \\ \text{false} & \text{otherwise} \end{cases} \quad (40)$$

For cases where resource limitation is relevant, X has to be further separated into a resource for aquatic and terrestrial stages, respectively.

The complete list of model parameters is given in Table 1. Each parameter may additionally be life-stage specific. In this case, a corresponding superscript indicates the life stage at which the switch between values occurs.

2.1.3. Incorporating individual variability in growth and metamorphosis timing

The time to reach metamorphosis can vary widely among individuals of the same species. Such individual variability can in turn have an effect on the projected system dynamics, for example when a model is applied to the population level. It can also introduce sampling biases in averaged growth trajectories, which are often used for DEB model calibration. We explicitly included individual variability in the analysis by simulating multiple individuals for each parameter set and returning the averaged growth trajectories.

Individual variability was introduced using the zoom factor Z , which was drawn from a truncated Normal distribution with mean 1, limits $(0, Inf)$ and standard deviation $\sigma_{Z,M}$.

The zoom factor is defined as the ratio between the maximum body sizes of two organisms, traditionally structural length (Kooijman et al., 2008). Since the present study uses dry mass as model currency, it makes sense to express the zoom factor on the basis of structural mass, rather than length, yielding the mass-based zoom factor Z :

$$Z = \frac{S_{max}}{S_{max,ref}} \quad (41)$$

$S_{max,ref}$ is the maximum structural mass of the reference organism. In the context of the present analysis, the reference is the population mean and Z reflects the deviation from the population mean. Based on the equilibrium conditions of the model, the zoom factor primarily propagates to the maximum size-specific ingestion rate $\{I\}_{max}$ to match S_{max} and other parameters which are assumed to scale with size:

$$\{I\}_{max} = Z \{I\}_{max,ref}^{1/3} \quad (42)$$

$$X_{int}^{emb} = X_{int,ref}^{emb} Z \quad (43)$$

$$K_X = K_{X,ref} Z \quad (44)$$

$$H^p = H_{ref}^p Z \quad (45)$$

A propagation of the zoom factor to H^j was omitted because this would cancel out the variability in the time to reach metamorphosis.

Otherwise, this configuration is based on other DEB modelling studies that incorporated individual variability, e.g. by Martin et al. (2012). Using this configuration, we can simulate a pattern where individuals differ in size, time to reach metamorphosis and size at the beginning and completion of metamorphosis (Fig. 1).

2.2. Global sensitivity analysis

One of the goals of this study was to develop a model that can reproduce a variety of patterns regarding metamorphosis traits, including varying lengths of metamorphosis constant dry mass at metamorphosis. Furthermore, the individual's ability to complete metamorphosis, once metamorphic climax has been reached, should be fairly insensitive to the choice of model parameters.

Due to the dependency of the metamorph life stage on larval parameters, it is not immediately obvious how changes in parameters result in changes in life history-traits. Therefore, we performed a global sensitivity analysis using the Sobol method (Sobol, 2001), focusing on inferred traits which can also be observed in life history-experiments. These were the growth rate during the larval period, the duration of the larval period, the duration of metamorphosis and the relative weight change during metamorphosis.

When expressing all life-history traits in absolute terms, it is to be expected that the size-specific maximum ingestion rate $\{I\}_{max}$ is most influential on all traits. Therefore, the duration of metamorphosis

Table 1

Species-specific model parameters for the dynamic energy budget anuran model. Note that parameters can in addition be life stage-specific, indicated by optional superscripts.

Parameter	Unit	Description
Z	–	Mass-based zoom factor
X_{int}^{emb}	mg	Initial mass of vitellus, approximated as dry mass of an egg
K_X	mg L ⁻¹	Half saturation for food uptake
$\{i_{max}\}$	mg mg ^{-2/3} d ⁻¹	Surface area-specific maximum ingestion rate
η_{TA}	–	Assimilation efficiency
η_{AS}	–	Growth efficiency
η_{AR}	–	Reproduction efficiency
κ	–	Allocation fraction to soma
γ	–	Allocation to metamorphic reserves
k_M	d ⁻¹	Somatic maintenance rate constant
k_J	d ⁻¹	Maturity maintenance rate constant
H^J	mg	Maturity at metamorphosis (approximately GS 42)
H^P	mg	Maturity at puberty

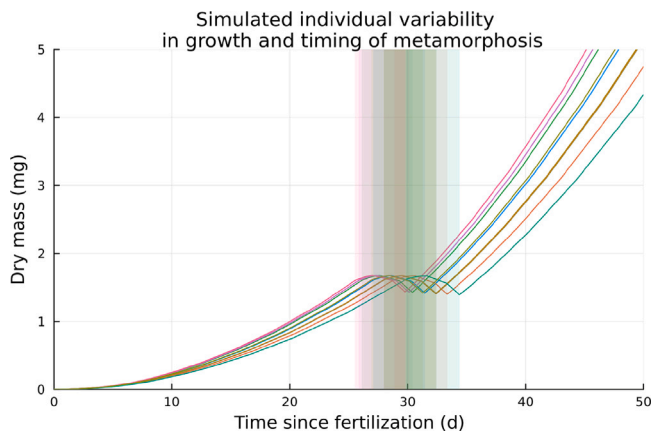


Fig. 1. Example simulation of larval and metamorph growth with account for individual variability. Each line represents an individual, shaded areas indicate the duration of metamorphosis for each individual.

was expressed relative to the duration of the larval period, in order to reduce the correlation in this output with other life history-traits.

The relative weight change was expressed in dry mass, since this is the central model currency and more relevant to energy budgets than wet mass. Based on a total of $16 \cdot 10^3$ Sobol samples, we produced two groups of outputs: The first output were the first-order and total order sensitivity coefficients, including their 95% confidence intervals. The first-order sensitivity indices describe the sensitivity of model outputs when considering variation of individual parameters, whereas the total-order sensitivity indices take interactions with other parameters into account. The second output was a summary of the trait combinations, i.e. the probability distributions of life-history traits and their correlations.

2.3. Parameter inference

For parameter inference, we used Population Monte Carlo Approximate Bayesian Computation (PMC-ABC, Beaumont et al. (2009)). PMC-ABC is not a widely used method within the context of DEB modelling, but well-suited for parameter inference in dynamic models with stochastic output in general (Beaumont, 2010). Conceptually, the working principle of PMC-ABC is that parameter combinations are sampled from the priors, the data is simulated and the distance between data and simulations is calculated. Based on the distance values, parameter combinations are either accepted or rejected and accepted parameters are re-sampled with perturbation and sampling weights. The sampling weights are defined so that, by repeating the process of re-sampling, simulation and rejection, the resulting weighted

distribution of accepted parameter combinations approximates the theoretical joint posterior distribution. As distance function, we used the Euclidean distance, which is a standard choice in ABC (Prangle, 2017). All response variables were scaled by their respective maxima and $\ln(x + 1)$ -transformed. PMC-ABC is a practical choice for this fitting problem because the model output is stochastic, and PMC-ABC is well-suited to deal with stochasticity in model outputs. PMC-ABC does furthermore not require any gradients, which avoids issues due to discontinuities in the simulation.

The parameter inference is documented in detail in the supporting information.

2.3.1. Inference of *D. galganoi* larval parameters

To infer larval parameters from data, we used data on larval growth and metamorphosis in *Discoglossus galganoi*. The data included average time-resolved wet mass of tadpoles (< Gosner stage 42), the time-resolved fraction of non-metamorphosed survivors as well as wet mass and time since birth at Gosner stages 42 and G46, respectively (corresponding to the control data generated and previously analyzed by González-López et al. (2026)).

The prior means for the surface area-specific ingestion rate for larvae ($\{i_{max}^{lv}\}$) and somatic maintenance rate constant for embryos, larvae and metamorphs (k_M^{emb}) were adjusted to approximate the maximum size of tadpoles up to an order of magnitude prior to the application of PMC-ABC. For details on calibration settings, see Supporting Information. Apart from the primary DEB parameters, it was necessary to add estimates for the water contents of larvae and freshly emerged juveniles, respectively, because the water contents at the beginning and end of metamorphic climax can have a large effect on the implied change in dry mass. We defined specific priors for water contents based on empirical information, consisting of 12 measurements of water content in *D. galganoi* juveniles and 5 pooled measurements of water content in *P. perezi* larvae (10 larvae per measurement). For larval water content, we defined the prior as a Normal distribution with the mean and standard deviation equal to their empirical values. For the juvenile water content, we used twice the empirical standard deviation, since we expected water content to be more variable for terrestrial life stages, and the empirical distribution stems from a different species. In order to be able to match the first mass measurement, we included the auxiliary parameter *time since birth*, referring the time since birth (onset of feeding) at which the first measurement in the experiment was taken. The dry mass of eggs, which is also equal to the initial mass of simulated individuals, was set to a measured value of 1 mg.

2.4. Inference of *D. galganoi* juvenile and adult parameters

In a second step, we fitted the model to an observation of the clutch size and known female mass, the maximum wet mass of adults (Oliveira et al., 2017) and the juvenile relative growth rate (observed in the closely related *D. pictus*, Martel et al. (2025)). These observations

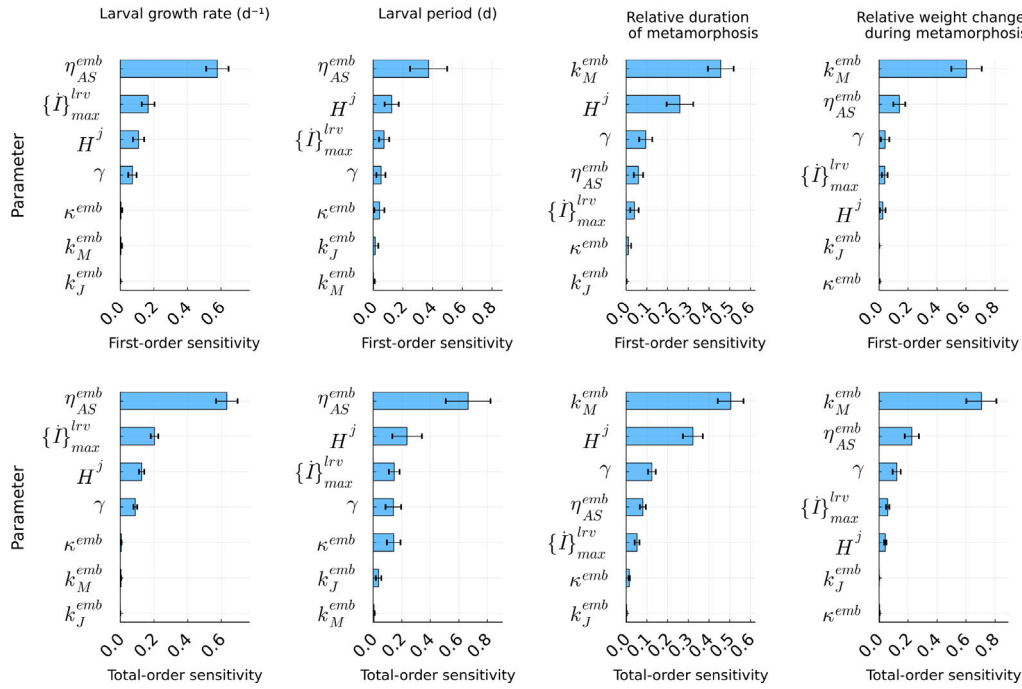


Fig. 2. First-order and total-order sensitivity indices with respect to larval and metamorph developmental traits. Error bars show 95% confidence intervals. Parameter superscripts refer to the first life-stage at which a life stage-specific value is valid (*emb* = embryo, *lrv* = larva).

were used to infer estimates for a subset of the juvenile and adult parameters, in order to emulate a case where fewer information is available, and therefore fewer information is available. The priors for these parameters were set manually to match the observations up to an order of magnitude. With respect to life stage-specificity of parameters, we assumed that terrestrial stages differ from aquatic stages in their maximum size-specific ingestion rate $\{\dot{I}\}_{max}^{juv}$, growth efficiency η_{AS}^{juv} and maturity maintenance rate k_J^{juv} .

Furthermore, the reproduction efficiency η_{AR} and maturity threshold for puberty, H^p , were estimated in order to match the clutch size at specific female mass. This choice was motivated by the implications of these parameters for the observed life history traits: The maximum size-specific ingestion rate has a large influence on the maximum size, and is therefore used to correct for the size-difference between frogs and tadpoles. It is also biologically plausible to adjust this parameter since frogs and tadpoles have different diets. The growth efficiency was adjusted because this is the only parameter which directly affects the somatic growth rate without interfering with maximum size. The maturity maintenance rate constant was adjusted to allow for a realistic age at puberty, which was not possible by adjusting H^p alone.

The prior modes for juvenile and adult parameters were adjusted manually and assigned a standard deviation corresponding to a 10% coefficient of variation. A prior predictive check was conducted to test verify that the prior distributions do not inhibit a good fit to the data.

The assimilation efficiency η_{IA} was fixed to a default value of 0.54 (Pfab et al., 2020). The initial egg dry mass X_{int}^{emb} was set to 1 mg based on the average measured value in 25 eggs.

2.4.1. Inference of *P. perezi* larval parameters and cross-validation

The dataset which was available for *P. perezi* had the same structure as the dataset used for inference of *D. galganoi* larval parameters. For most anuran species however, the available datasets will be less detailed. Therefore, we used this dataset for a cross-validation exercise, in order to check how well traits related to Gosner stage 46 can be predicted from observations up to Gosner stage 42. We calibrated the *P. perezi* model only to data of larval growth and the average time to reach Gosner stage 42, omitting the time-resolved number of tadpoles,

time to reach Gosner 46 and wet mass at Gosner 46. We used the point estimates derived for *D. galganoi* as defaults and estimated a subset of parameters for *P. perezi*, namely the mean zoom factor $\bar{Z}_{P,D}$ (size correction for *P. perezi* relative to *D. galganoi*), growth efficiency η_{AS}^{emb} , metamorphosis threshold H^j and time since birth at initial mass measurement. Based on this estimation, we simulated the entire dataset, thus attempting to predict the time-resolved number of tadpoles, wet mass at Gosner stage 46 and time to reach Gosner stage 46.

3. Results

3.1. Global sensitivity analysis

First-order sensitivity indices indicated the growth efficiency η_{AS}^{emb} as a highly influential parameter for all considered traits. As expected based on the model formulation, η_{AS}^{emb} was most influential on the larval growth rate, since this is the only parameter which affects the growth rate without simultaneously affecting maximum size. In combination with the maturity threshold H^j , η_{AS}^{emb} strongly influences the duration of the larval period (Fig. 2). The relative duration and relative weight change during metamorphosis were most sensitive to the maintenance rate constant k_M^{emb} . Among the considered traits, the newly introduced γ -parameter was most influential on the relative duration of metamorphosis and relative weight change during metamorphosis, based on its first-order as well as second-order sensitivity index. Total-order sensitivity coefficients indicated the same trends as first-order sensitivity coefficients with respect to the most influential parameters (Fig. 2), that is, a strong influence of the growth efficiency η_{AS}^{emb} on larval growth rate and larval period and a strong influence of the somatic maintenance rate constant k_M^{emb} on relative duration of metamorphosis and relative weight change during metamorphosis.

The allocation fraction to metamorphic reserves, γ , was generally not among the most influential parameters with respect to metamorphosis traits.

As expected, the larval growth rate and duration of the larval period were strongly negatively correlated in the global sensitivity analysis output (Fig. 3). A higher larval growth rate was also associated with a

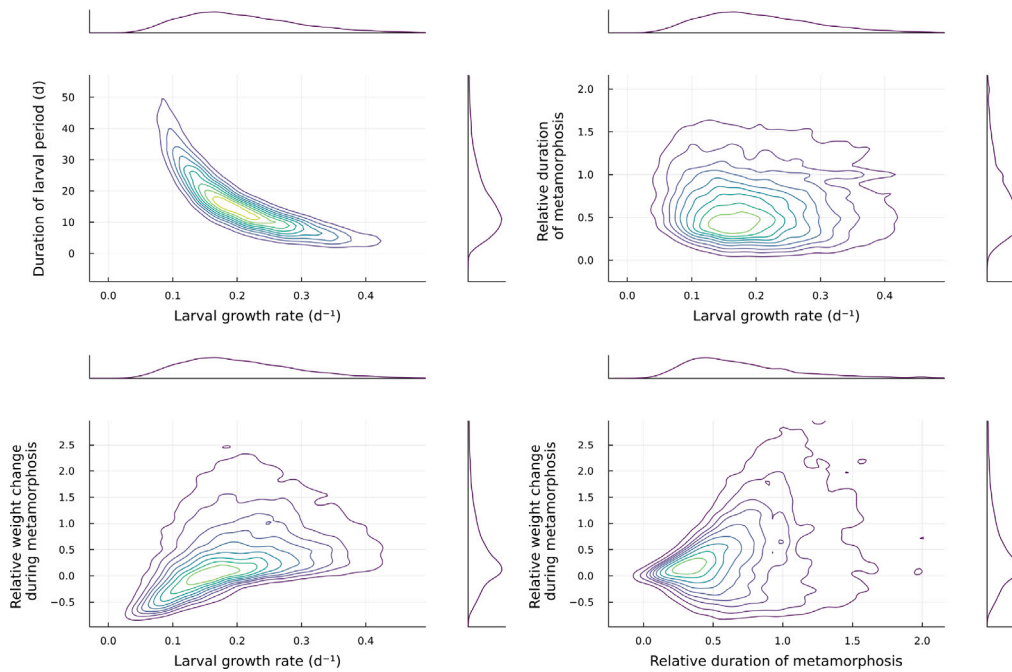


Fig. 3. Probability densities of binary trait combinations, based on evaluation of 16,000 Sobol samples. Brighter colours indicate higher probability density. Relative weight change is expressed in dry mass.

higher relative weight change during metamorphosis. With respect to relative weight change and duration of metamorphosis, the most likely outcome was constant dry mass in combination with a metamorphosis which was ca. 50% shorter than the larval period (relative duration of metamorphosis < 1). In cases where metamorphosis was longer than the larval period (relative duration of metamorphosis > 1), an increase in dry mass, constant dry mass or decrease in dry mass was possible (Fig. 3). While it is possible to construct this case, it is unlikely to be relevant for real amphibian species.

3.2. Inference of larval parameters for *D. galganoi*

The larval mass data for *D. galganoi* showed an initial growth phase followed by a decline in the average wet mass as larvae approached metamorphosis (Fig. 4). The end of the growth phase and start of the decline coincided with the first larvae entering metamorphosis, and was therefore likely an artefact of the measured mass being the average across ten larvae per aquarium. That means, as the fraction of metamorphosed individuals increased, this introduced a bias towards smaller individuals, leading to a decrease in the average mass, while each individual likely followed a monotonic positive growth trajectory during the larval period.

This could be accounted for by explicitly simulating individual variability during model calibration, and using the fraction of non-metamorphosed larvae (*Number of tadpoles*). The calibrated model reproduced this pattern, and simultaneously matched the time and wet mass associated with Gosner stages 42 and 46 (Fig. 4).

The estimated parameter values are reported in Table 2 and the marginal posterior distributions are shown in Fig. 5. For most parameters, the marginal posterior distributions were clearly narrower than the priors. Remarkably, this was also the case for the width of the individual variability kernel $\sigma_{Z,M}$, which suggests that the larval growth data and time-resolved probability of reaching metamorphosis were informative on the individual variability in metabolic rates. Judging from the marginal posterior, the data was also informative on the newly introduced γ parameter. However, not all parameters could be unequivocally estimated from the data. Relative to the prior distributions, marginal posteriors were particularly wide for the larval growth

Table 2

Parameter estimates for *D. galganoi* larvae and metamorphs. *Time since birth* refers to the time since birth at the first larval wet mass measurement. Median refers to the median of the posterior distribution. P_{05} and P_{95} are the corresponding percentiles of the marginal posterior.

Parameter	Best fit	Median	P_{05}	P_{95}
$\sigma_{Z,M}$	0.153	0.252	0.16	0.304
$\{i\}_{AS}^{lrv}$	1.79	1.61	1.38	1.86
k_M^{emb}	0.213	0.175	0.135	0.242
η_{AS}^{emb}	0.78	0.878	0.62	0.979
H^j	9.11	12.3	7.01	18.9
γ	0.829	0.718	0.564	0.828
k_J^{emb}	0.0088	0.0117	0.00104	0.0408
κ	0.81	0.739	0.67	0.857
<i>Larval water content</i>	0.935	0.94	0.934	0.946
<i>Juvenile water content</i>	0.892	0.864	0.792	0.906
<i>Time since birth</i>	23.5	20.6	16.7	24.3

efficiency η_{AS}^{lrv} . For the ingestion rate $\{i\}_{AS}^{lrv}$, we had constructed a specific prior, resulting in a relatively small difference between prior and posterior.

The auxiliary parameters *larval water content*, *juvenile water content* and *time since birth* could, perhaps unsurprisingly, also not be unequivocally inferred from the data. However, the estimate for the juvenile water content was distinctly higher than the empirical prior mean (Fig. 5). A lower estimate likely would have resulted in an over-estimation of the change in wet mass during metamorphosis.

3.3. Inference of juvenile and adult parameters for *D. galganoi*

The model could approximate post-metamorphic relative weight change, maximum wet mass and weight-specific clutch size (Fig. 6), while also implying a considerable amount of individual variability in these traits, resulting from the individual variability parameter σ_Z which had been estimated from larval data.

The implied age at maturity was less than one year, clearly lower than the reported value of 4 years (Oliveira et al., 2017). This might

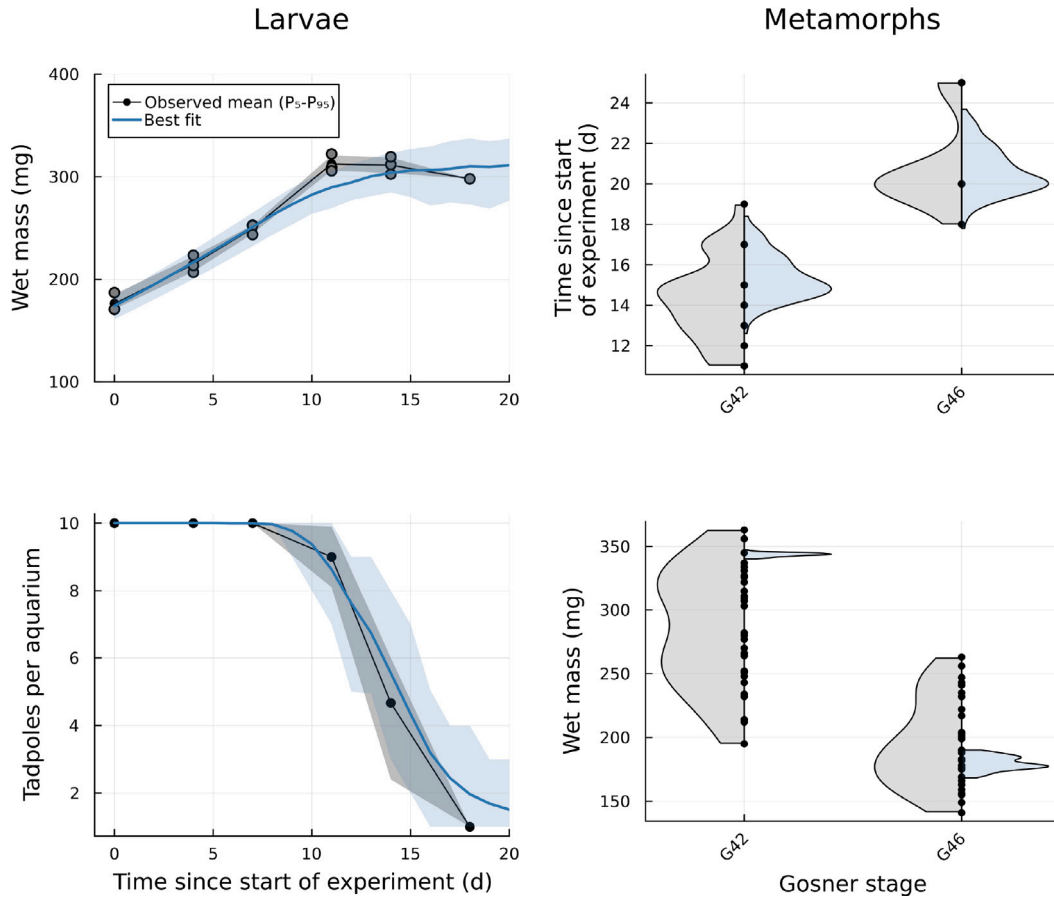


Fig. 4. Model fit to larval and metamorph *Discoglossus galganoi* data. Simulated values show the distribution of 100 evaluations of the best fit. The variability in simulated values results from simulated individual variability. Left: Fit to larval growth data. Lines show averages, shaded areas the 5th to 95th percentile. Right: Fit to metamorph data. Shaded areas show kernel density estimates. Grey areas show the variability in the data with dots observed values per individual as dots. Blue areas show the variability in the fitted mean value.

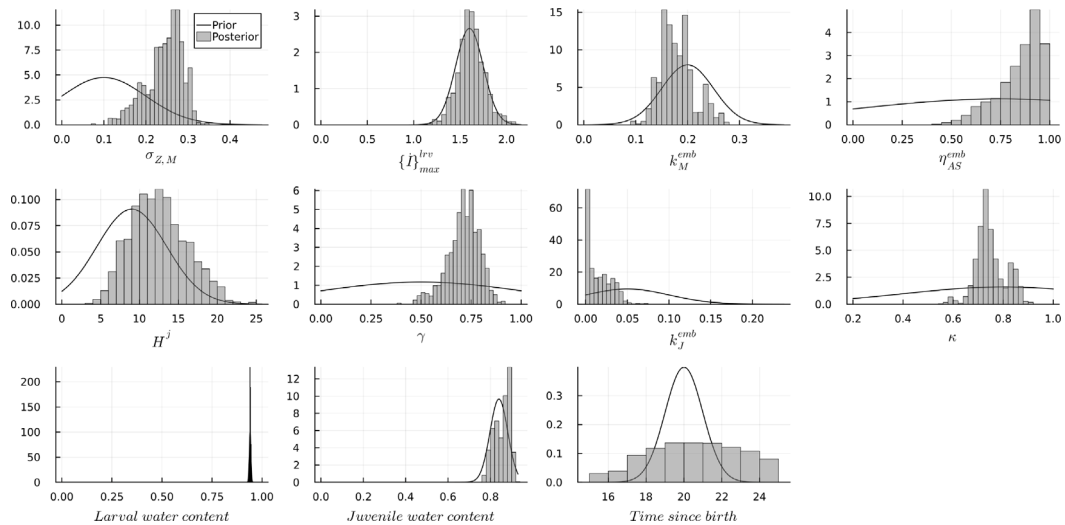


Fig. 5. Marginal posterior distributions of parameters for *D. galganoi* larvae and metamorphs. Vertical axis shows probability density.

be a result of simulating constant environmental conditions (Fig. 7), neglecting factors such as seasonal temperature fluctuations and hibernation. The calibration procedure did not allow for a full characterization of the posterior distribution due to very low acceptance rates when fitting to the juvenile and adult data. The implied credible intervals are not reliable, and we only report point estimates (Table 3).

3.4. Inference of larval parameters for *P. perezi* and cross-validation

In *P. perezi*, larval growth and metamorphosis traits at Gosner stage 42 were reproduced well by fitting a subset of parameters (Table 4) to this data, and leaving the remaining parameters at the point estimates for *D. galganoi*. The time to reach Gosner stage 46 and wet mass at

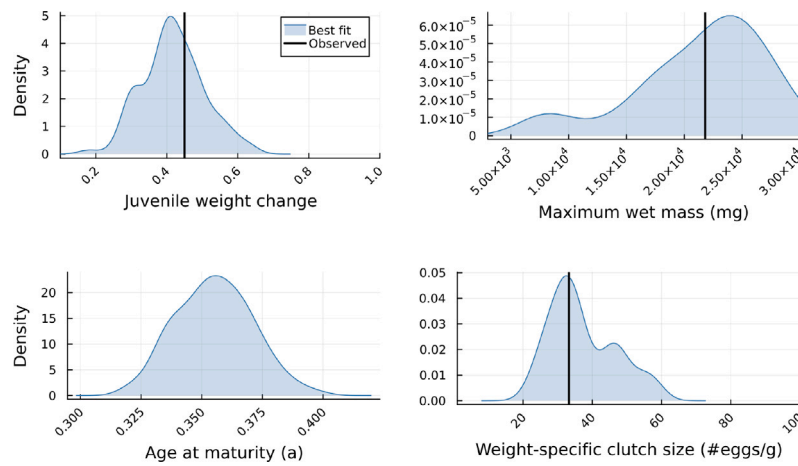


Fig. 6. Simulated life-history traits for *D. galganoi* juveniles and adults after calibration with observed traits. Variability in simulation output results from simulated individual variability, not parameter uncertainty. The age at maturity is the theoretical value obtained for constant temperature.

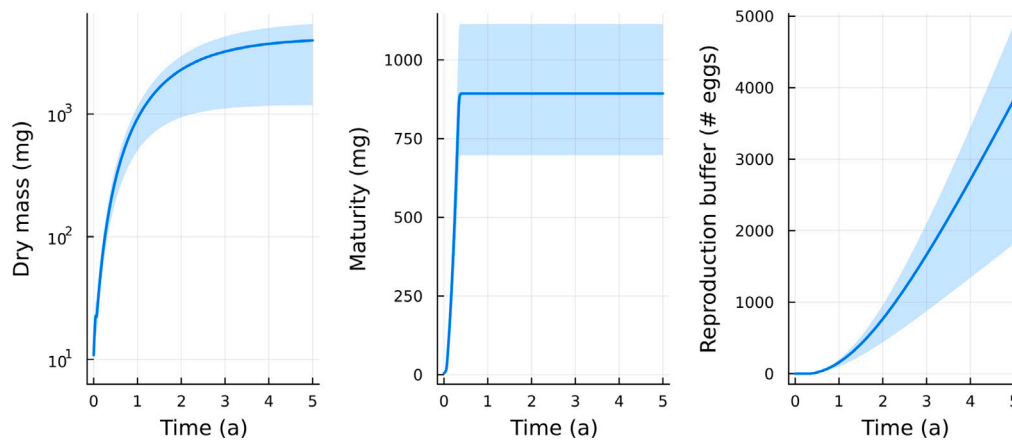


Fig. 7. Simulated growth and reproductive investment of *D. galganoi* under constant conditions. The simulation ignores any effects of seasonal changes, for example in temperature, leading to an earlier age at maturity than expected in reality. Error bands show the 5th to 95th percentiles resulting from individual variability.

Table 3

Parameter estimates for *D. galganoi* juveniles and adults.

Parameter	Best fit
$\{I\}_{max}^{juv}$	6.09
η_{AS}^{juv}	0.043
H^p	889
κ_J^{juv}	0.00822
η_{AR}	0.0191

Table 4

Estimates of *P. perezii* larval parameters. Median refers to the median of the posterior distribution. P_{05} and P_{95} are the corresponding percentiles of the marginal posterior.

Parameter	Best fit	Median	P_{05}	P_{95}
$Z_{P,D}$	1.32	1.4	1.1	2.24
η_{AS}^{emb}	0.631	0.609	0.346	0.839
H^j	68.5	117	67.4	168
Time since birth	54.4	45.3	28.9	67.6

Gosner stage 46 were under-predicted with a mean relative percent error of 16% and 28%, respectively (Fig. 8).

4. Discussion

4.1. Model structure: Advantages, weak points and future refinements

We presented a novel DEB model formulation for amphibians, which, among previous DEB model formulations, is most closely related to the model for amphibians previously developed by Pfab et al. (2020).

Compared to the stdDEB approach (Romoli et al., 2025), the presented model has one parameter less. Due to the application of DEBkiss (Jager et al., 2013), one abstract parameter (the maturity at birth) was replaced with a measurable parameter (the dry mass of eggs). By extending the DEBkiss logic for embryos to metamorphs, the reserve

allocation fraction γ effectively replaced the maturity threshold for the end of metamorphosis in the stdDEB model. Since maturity thresholds are unmeasurable quantities and inherently correlated with other parameters, reducing the number of maturity thresholds in the model should simplify parameter inference, in particular when global calibration methods are applied. During metamorphosis, our model departs from standard DEB assumptions so that the dynamics can be computed without the introduction of additional parameters. Romoli et al. (2025) achieved this by assuming homeostasis during metamorphosis, keeping the model in line with standard DEB assumptions throughout the entire life cycle.

A crucial difference compared to the existing DEBkiss approach for amphibians (Pfab et al., 2020) is the inclusion of maturity and maturity maintenance, which allows to decouple life stage transitions from body

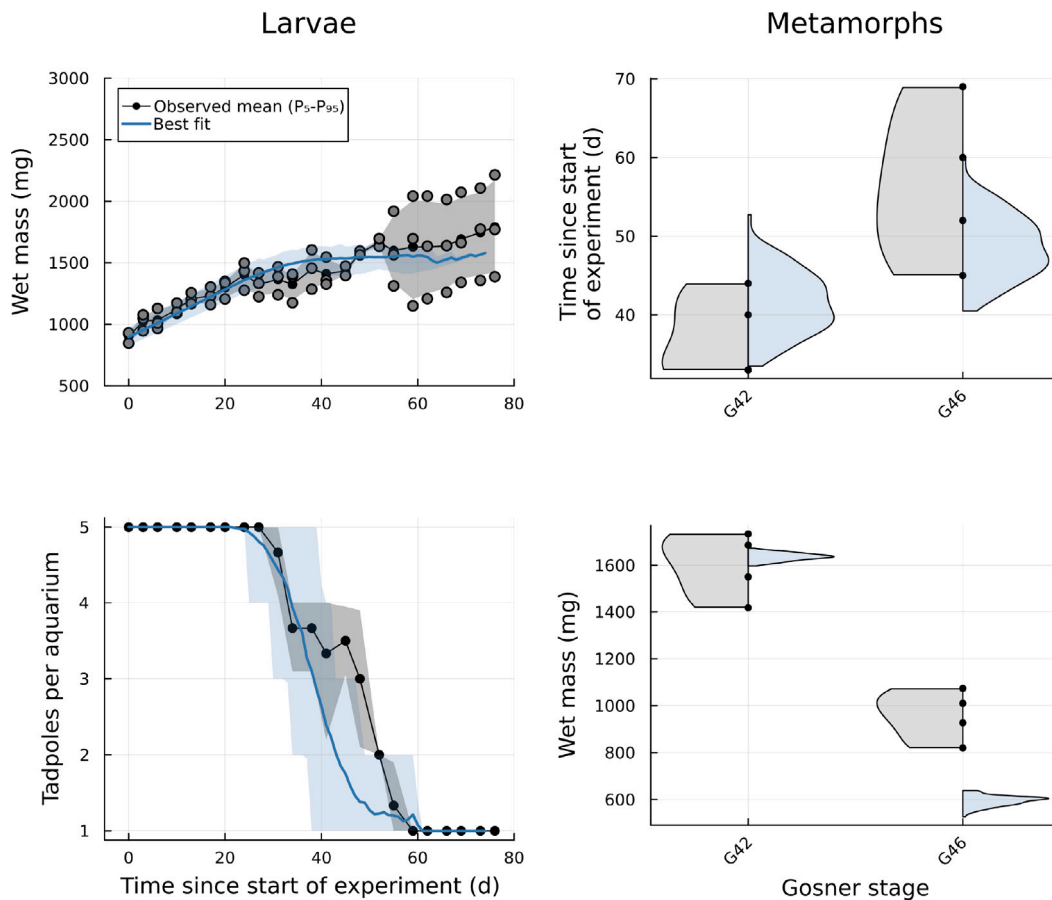


Fig. 8. Model fit to larval and metamorph *Pelophylax perezi* data. Simulated values show the distribution of 100 evaluations of the best fit. The variability in simulated values results from simulated individual variability, not parameter uncertainty. The model was only fitted to larval wet mass and metamorphosis traits at Gosner stage 42. Non-monotonicity in the average number of tadpoles is the result of a missing value. Left: Fit to larval growth data. Lines show averages, shaded areas the 5th to 95th percentile. Right: Fit to metamorph data. Shaded areas show kernel density estimates. Grey areas show the variability in the data with dots observed values per individual as dots. Blue areas show the variability in the fitted mean value.

size. This is generally relevant for amphibians, since they exhibit high plasticity in developmental rates in response to environmental conditions (Newman, 1992). Including both maturity and a dedicated reserve department necessitated an additional split in the resource allocation dedicated to the accumulation of reserves, rather than using the $1 - \kappa$ -branch as previously done for fish in a DEBkiss context (Martin et al., 2017). This allows to retain maturity as a separate compartment and results in higher model flexibility, with the penalty of one additional model parameter.

Another factor differentiating existing models is how food ingestion and assimilation during and before metamorphosis are treated. Pfab et al. (2020) modelled changes in ingestion rates in a particularly detailed way, by estimating a separate size-specific ingestion rate for each Gosner stage. This provides a detailed description of the change in ingestion rate during development. Another approach is to assume a constant size-specific ingestion rate during larval development, and let ingestion immediately drop to 0 during metamorphosis (Romoli et al., 2025). In the present study, we took an intermediate approach, assuming a constant size-specific ingestion rate during larval development and a gradual decrease during parts of metamorphosis, limiting the fasting period to a fraction of the time between Gosner stages 42 and 46. This was mostly crucial to allow for flexibility in the predicted relative dry mass changes during metamorphosis, as shown by the outcome of the sensitivity analysis. Since the gradual decrease in ingestion rates allows to retain more structure than a sudden drop to 0 would, this model can in principle also reproduce patterns of constant dry mass, as demonstrated by the global sensitivity analysis.

However, the fact that our model allows for continued feeding during metamorphosis is a weak point of the current model formulation, as the assumption of a sharp drop in ingestion rate with the onset of metamorphic climax is more biologically plausible. Even with the addition of a gradually changing ingestion rate, the model tended towards over-predicting mass loss during metamorphosis, as was evident in the cross-validation study with *P. perezi*, and indirectly so in a relatively high implied water content of juveniles in *D. galganoi*. Therefore, efforts to improve the biological realism of DEBkiss-based amphibian models may focus on implementing stricter rules for the fasting period during metamorphosis, while maintaining the fit to metamorphic weight change data. One approach could allow for different energy contents between compartments, since a higher energetic content of E^{mt} would lead a lower mass reduction during metamorphosis. This would also require an additional parameter to estimate, the energy density of the metamorphic reserve, relative to that of the remaining masses. If necessary however, it might be possible to derive an informative prior from experimental data (Sheridan and Kao, 1998).

In addition, we made the assumption that E^{mt} is not associated with any maintenance costs. Since we defined E^{mt} to be the biomass that is consumed during metamorphic climax, this assumption is a simplification of biological reality, where some of the biomass that is consumed, such as the tail, does in fact require maintenance. To weaken this assumption, one might consider to introduce maintenance costs for E^{mt} as well. This, too, requires that the consequences for parameter estimation are tested.

4.2. Accounting for individual variability

In order to accurately reproduce the trajectory of average body mass during larval development, we included an explicit representation of individual variability. The model explained the trajectory of average larval body mass through an observation bias that arises from larger individuals reaching metamorphosis earlier, leading to a decrease in average mass towards the end of the larval period, while the growth trajectory of the simulated individuals remained monotonically increasing. Such a representation of individual variability is not only relevant for DEB parameter inference, but can also have consequences for understanding the relationship between amphibian energy budgets and effects of environmental factors on life-history. For example, Székely et al. (2020) demonstrated a relationship between size at metamorphosis and survival probability after metamorphosis. Such relationships could be incorporated into the simulation of individual variability as presented in the present study, since it implies individual variability in both the timing of metamorphosis and the size at metamorphosis. The time to reach metamorphosis is an important endpoint in the analysis of chemical toxicity to amphibians, and therefore an accurate representation of this endpoint in the mechanistic and statistical model is a relevant step towards routinely analysing amphibian toxicity data with mechanistic models.

In cases where the observed timing of metamorphosis may be less variable, the individual variability in timing of metamorphosis could be eliminated by propagating the zoom factor to the maturity threshold for metamorphosis, leading to variability in growth, but not in the time to reach metamorphosis.

4.3. Fitting the model to whole life-cycle data

The calibration for the whole life cycle of *D. galganoi* was split into one calibration to larva/metamorph data and one to the juvenile/adult data. Still then, the characterization of uncertainties for juveniles/adult parameters was computationally demanding. The underlying challenge is that we attempted to perform a global parameter search with stochastic model output where some parameters are inherently correlated. This challenge would be amplified when fitting to all data simultaneously, which is desirable, but did not turn out to be feasible in the current setting.

The most straightforward way to simplify the matter is to ignore individual variability, so that the model output becomes deterministic again, and apply global optimization or standard Bayesian inference. For routine applications of the model to small datasets, we suggest that it is probably the most practical approach. However, one should then be aware of possible sampling biases when using averaged growth data. Furthermore, unless we resort to local optimization, the correlation between juvenile/adult and larval/metamorph parameters will still be a challenge.

Furthermore, there are approaches which allow to apply a divide-and-conquer strategy while still inferring the joint posterior of all parameters, such as Gaussian copula ABC (Li et al., 2017). We are not aware of any applications of these methods to DEB models, nor of software packages which allow for their immediate application, but exploring this path might prove useful for calibration of DEB models across multiple life stages.

4.4. Conclusions

A Dynamic Energy Budget Model for amphibians was derived from DEBkiss and extended to include a reserve compartment, which acts as a dedicated storage compartment for metamorphosis. This model formulation was targeted towards introducing not more parameters than necessary while providing flexibility in the reproducible metamorphosis traits, as was confirmed by a global sensitivity analysis. While we could demonstrate that a simplified model formulation can be used to

include a metamorphic reserve compartment in a model derived from DEBkiss, open questions remain regarding the implementation of bioenergetics during metamorphic climax in such a simplified, DEBkiss-based model. First and foremost, these concern the alignment of biologically realistic reductions in feeding rates with observed changes in body mass during metamorphosis, as well as the contribution of different tissues and functional compartments to the fuelling of maintenance, maturation and recycling of structure. The model also described life-history of two anurans, including a representation of the whole life cycle in *D. galganoi*. By explicitly simulating individual variability, the model could be fitted to averaged time-resolved growth data without falling victim to a sampling bias due to larger individuals reaching metamorphosis earlier. The model can also be calibrated to a reduced dataset, including only data up to Gosner stage 42. Time to reach Gosner stage 46 and wet mass at Gosner stage 46 were under-predicted, but within a reasonable margin of error. After having applied the model to two distinct species of anurans, we can envision an application of the model to urodela and possibly reptiles, given the availability of suitable life-history data. Simultaneous estimation of all model parameters, with account for individual variability, parameter uncertainty and in the context of a global parameter search, represents a computational and methodological challenge which could incite the exploration of novel calibration techniques for DEB models.

CRedit authorship contribution statement

Simon Hansul: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation. **Samuel González-López:** Writing – review & editing, Investigation, Data curation. **Manuel E. Ortiz-Santaliestra:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition. **Jean Lou C.M. Dorne:** Writing – review & editing, Resources, Project administration. **Andreas Focks:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ecolmodel.2026.111715>.

Data availability

Data will be made available on request.

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